MENOFYIA UNIVERSITY

SYSTEM ANALYSIS & DESIGN PROJECT

SECOND YEAR

2026-2027

جامعة المنوفية

كلية الحاسبات والمعلومات

2026-2027

Water Supply Complaint & Billing System

No.	Name	Section
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Planning (Project Plan)

Project Identification

Project Name:

Water Supply Complaint & Billing System

Current Problem:

1- Citizens have to visit the company office or call to report a water complaint, and it is hard to follow up on it.
2- Bills are often based on estimated readings, which causes disputes and overcharging.
3- Water losses and leaks are reported late, so a large share of produced water is never paid for.
4- Complaint and billing records are paper-based or scattered, with no central data to analyze.

Description about project:

- The system lets a customer register a complaint (no water, low pressure, leak, wrong bill, meter fault) online or by phone, and automatically sends it to the right technical team.
- Each complaint gets a tracking number, so the customer can follow its status until it is closed.
- Duplicate complaints from the same area are merged into one incident.
- The system stores meter readings, calculates the bill from actual consumption, and shows the customer their reading history.
- Customers can receive e-bills and SMS reminders, and pay online.
- Managers get reports on complaints, leaks, response times and collection. This data can later be used for data analysis and ML (not implemented now).

System Request

System Request— Water Supply Complaint & Billing System	
Project Sponsor:	Water & Sanitation Company, Customer Service and Commercial Department (e.g. Menoufia branch)
Business Need:	1- Reduce billing errors and disputes. 2- Reduce water loss by reporting leaks faster. 3- Make complaint submission available to everyone, online. 4- Track every complaint until it is solved.
Business Requirements:	1- Online and phone complaint registration with tracking number. 2- Meter reading and bill calculation module. 3- Online payment and SMS/e-bill notifications. 4- Reports and dashboard for management.
Business Value:	1- Providing accurate information. 2- Saving time. 3- Saving cost. 4- Saving effort and water.
Special Issues or Constraints:	1- Deadline in 1 January. 2- Must integrate with existing customer and meter data. 3- Customer data must be protected.

Feasibility Analysis

Technical Feasibility

1. The system can be built as a web application using HTML, CSS, JavaScript and a back-end such as Java or Python, with a relational database.
2. Development tools (VS Code, IntelliJ, Git, Jira) are free or already available to the team.
3. It needs an internet connection and works on any browser, on Windows or Android.
4. SMS and online payment gateways can be added through standard APIs. These services already exist in Egypt's water sector.

Economic Feasibility

1- Estimated development cost is about **975,700 EGP** (including a 10% contingency).
2- Estimated operating cost is about **588,355 EGP per year**.
3- Estimated benefits are about **3,295,000 EGP per year** at full adoption (fewer disputes, less staff time, fewer field visits, savings on paper bills, better collection).
4- The 5-year analysis at a 15% discount rate gives an NPV ≈ 6.16 million EGP and a benefit-cost ratio ≈ 2.9, with payback within the first year (about 0.7 years).

Organizational Feasibility

Organizational feasibility assesses:

1. Management support: the water company is already moving to e-services (online complaints, payment, mobile apps), so management is likely to support the project.
2. Resource sufficiency: it needs customer-service staff, a technical team for field visits, and IT support. Staff need short training on the new system.
3. User acceptance: customers already use online bill payment, so a complaint-tracking portal fits existing habits.
4. Risk: resistance to changing paper procedures, which training and a phased rollout (one branch first) can reduce.

Time Estimation

	Planning	Analysis	Design	Implementation
Standard percentages	15%	20%	35%	30%
Estimated time based on planning time	Actual: 11 days	16 days	25 days	22 days

Tasks Identification

Name of Task	System Request
Start Date	9/11/2020
End Date	12/11/2020
Person assigned to task	Joulnar Hussein, rawan
Priority	High
Resources Needed	Microsoft word
Estimated Time	4 hours
Actual Time	2hours

Name of Task	Feasibility Study
Start Date	12/11/2020
End Date	17/11/2020
Person assigned to task	Ahmed mahmoud
Priority	High
Resources Needed	Microsoft word
Estimated Time	8 hours
Actual Time	5 hours

Name of Task	Pert Chart, Grantt Chart
Start Date	17/11/2020
End Date	19/11/2020
Person assigned to task	Ali raafat
Priority	High
Resources Needed	Ms project
Estimated Time	9 hours
Actual Time	6 hours

Name of Task	Interview
Start Date	19/11/2020
End Date	24/11/2020
Person assigned to task	Joulnar Hussein, rawan
Priority	Medium
Resources Needed	Microsoft word
Estimated Time	16 hours
Actual Time	14 hours

Name of Task	Questionnaire
Start Date	19/11/2020
End Date	23/11/2020
Person assigned to task	Ahmed allam
Priority	medium
Resources Needed	Form drive
Estimated Time	5 hours
Actual Time	2 hours

Name of Task	Data Flow Diagram
Start Date	24/11/2020
End Date	29/11/2020
Person assigned to task	Ahmed Mahmoud, Ahmed allam
Priority	High
Resources Needed	Internet
Estimated Time	1 day
Actual Time	18 hours

Name of Task	Data Store
Start Date	29/11/2020
End Date	2/12/2020
Person assigned to task	Ahmed mahmoud
Priority	High
Resources Needed	Microsoft word
Estimated Time	1 day
Actual Time	17 hours

Name of Task	process specification
Start Date	29/11/2020
End Date	3/12/2020
Person assigned to task	Joulnar hussein
Priority	High
Resources Needed	Microsoft word
Estimated Time	1 day and 10 hours
Actual Time	19 hours

Name of Task	Entity Relationship
Start Date	2/12/2020
End Date	6/12/2020
Person assigned to task	rawan
Priority	High
Resources Needed	StarUML

Estimated Time	1 day and 10 hours
Actual Time	19 hours

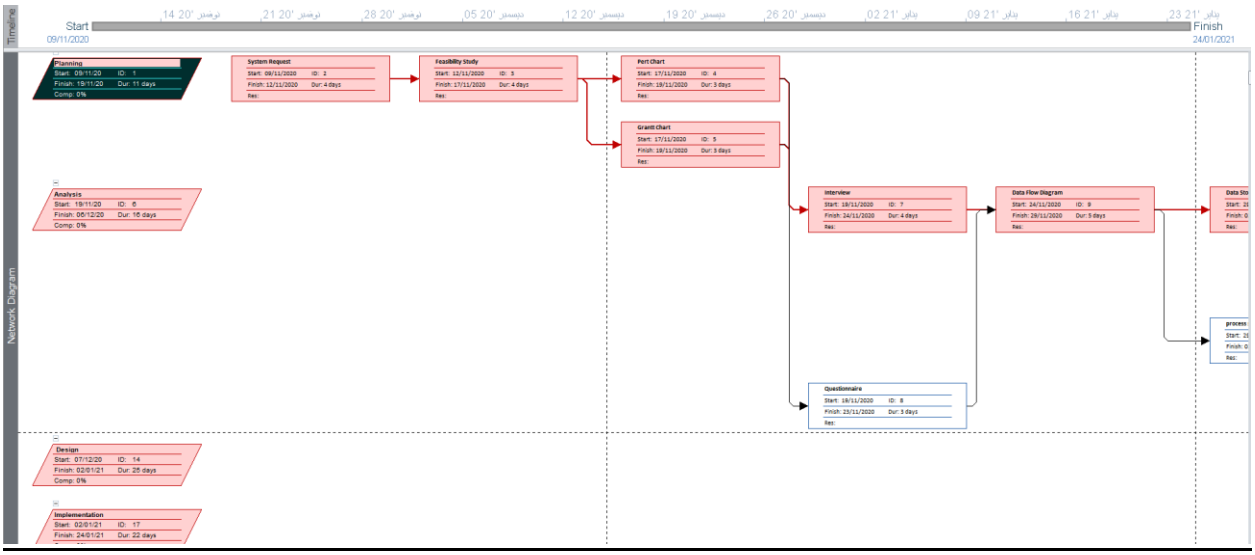
Name of Task	**Normalization**
Start Date	2/12/2020
End Date	6/12/2020
Person assigned to task	ali raafat
Priority	High
Resources Needed	StarUML
Estimated Time	1 day and 10 hours
Actual Time	19 hours

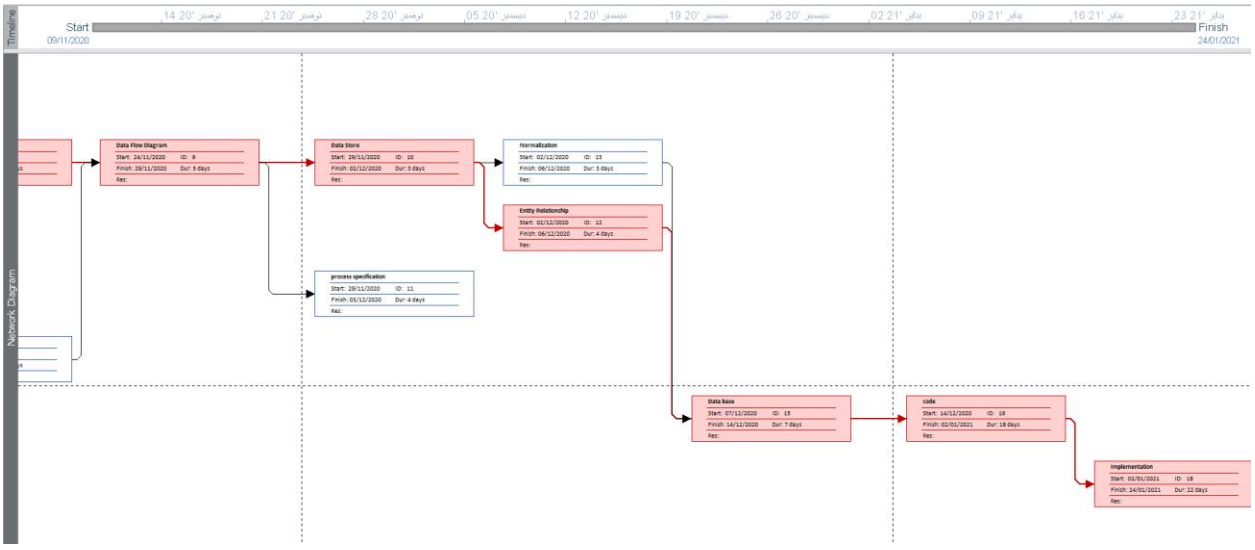
Name of Task	**design**
Start Date	7/12/2020
End Date	2/1/2021
Person assigned to task	-
Priority	High
Resources Needed	Visual studio code
Estimated Time	18 days
Actual Time	13 days

Name of Task	**Implementation**
Start Date	2/1/2021
End Date	24/1/2021
Person assigned to task	Ahmed allam ,ali raafat

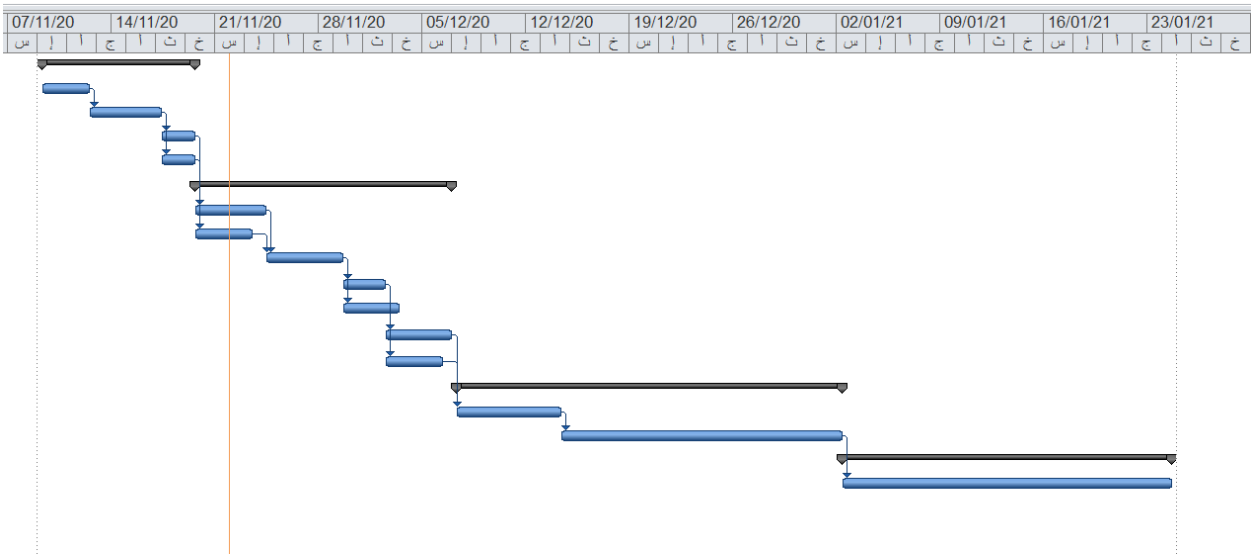
Priority	High
Resources Needed	Visual studio code
Estimated Time	16 days
Actual Time	11 days

Pert Chart (MS-Project)





Gantt chart (MS-Project)



Analysis (Project Proposal)

Interview

(include questions and answers)

1- What's your opinion of the current state of your business?

I think it's not good.

2- Is the system easy to use?

Yes, the system is simple

3- How does the system look like or work?

The system is a website can find the book that you want easily by entering the code or the name of the book, it enables the user to borrow any book he want, System works to arrange books in a specific way such as alphabetical order or by the book's authors name, the system can count the number of the books by the books code, less effort and without errors, it can be kept from being destroyed or lost. The system library contains all types of books in different fields.

4- What are the common errors your system have?

Loss of copyright to the authors

5- What are the critical objectives of your system?

- Critical objectives of my system
- Get profit
- Ease of use
- Accessing your large books and adding them on website

6- What technologies you use now in your current system?

- Low security system

7- What technologies you want to append to the new system?

- Make page on website for user registration
- Make page to Add a lot of books Not free
- Use multiple payment methods

8- If the user Find a problem can your system help him?

Yes, but system may can't solve other problem

9- Do you have anything else to add?

No, thanks

Interview Reports

Interview Report

Person Interviewed: Mr : Ahmed

Interviewer: Ahmed Allam

Purpose of Interview:

1. To see how the system works with it

2. Take advantage of its system and address errors

3. Knowing some of the system tools and how they will be developed

Summary of Interview:

- **The system work to show some free books in a website**
- **Common error is Loss of copyright to the authors**
- **He use Low security system**

Open Items:

- **1- Minimum 6 employees in one section.**
- **2- Create an application mobile to make the purchase process fast for the customer.**
- **3- Create a website page to make buying online easy to the user.**

Detailed Notes: See attached transcript.

Interview Report

Person Interviewed: mr ali

Interviewer: ahmed

Purpose of Interview:

1. **To see how the system works with it**
2. **Take advantage of its system and address errors**
3. **Knowing some of the system tools and how they will be developed**

Summary of Interview:

- **The system is a website can find the book that you want easily by entering the code or the name of the book, it enables the user to borrow any book he want**
- **Accessing your large books and adding them on website**
- **Make page on website for user registration**
- **Make page to Add a lot of books Not free**

Open Items:

- **1- Minimum 6 employees in one section.**
- **2- Create an application mobile to make the purchase process fast for the customer.**
- **3- Create a website page to make buying online easy to the user.**

Detailed Notes: See attached transcript.

Interview Report

Person Interviewed: eng mohamed

Interviewer: ahmed

Purpose of Interview:

1. **To see how the system works with it**
2. **Take advantage of its system and address errors**
3. **Knowing some of the system tools and how they will be developed**

Summary of Interview:

- **The system library contains all types of books in different fields and is a website.**
- **Common error is Loss of copyright to the authors**
- **Use multiple payment methods**

Open Items:

- **1- Minimum 6 employees in one section.**
- **2- Create an application mobile to make the purchase process fast for the customer.**
- **3- Create a website page to make buying online easy to the user.**

Detailed Notes: See attached transcript.

Questionnaire & RESPONSES

(using Google Forms, you must include questions and statistics)



System analysis project

Library Website System

*مطلوب

* ?do you prefer the system website or you suggest do the system application

- website
- application

* ?what's your opinion about the system

	5	4	3	2	1	
Need development	◯	◯	◯	◯	◯	Good

If the system needs to be developed, You can help us by adding some modifications to add to the ?system later

إجابتك

?If you are the user of your phone on our website, what is your satisfaction with the design

	5	4	3	2	1	
Need development	◯	◯	◯	◯	◯	Good

* ?" What is your satisfaction with the work of the website " system

	5	4	3	2	1	
not Satisfied	◯	◯	◯	◯	◯	Satisfied

* ?Is the system easy to use

	5	4	3	2	1	
No	○	○	○	○	○	Yes

* How easy to reach the books in System

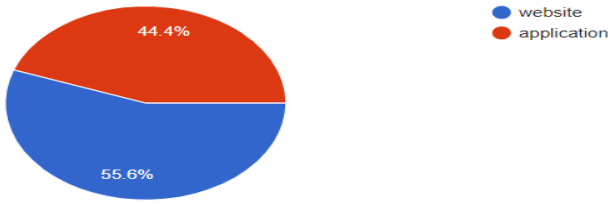
	5	4	3	2	1	
not easy	○	○	○	○	○	easy

?What are the common mistakes in the system

إجابتك

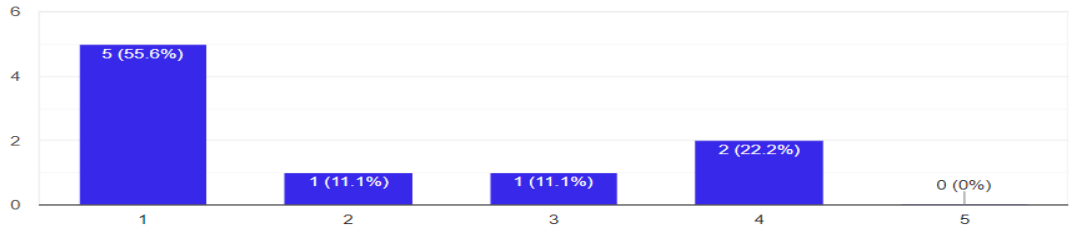
?do you prefer the system website or you suggest do the system application

9 ردود



?what's your opinion about the system

9 ردود



If the system needs to be developed, You can help us by adding some modifications to add to the system later?

4 ردود

No

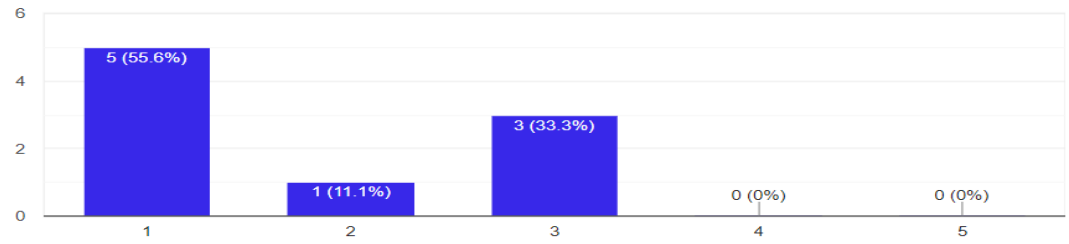
that need add some tools

No it doesn't need any develop

لا

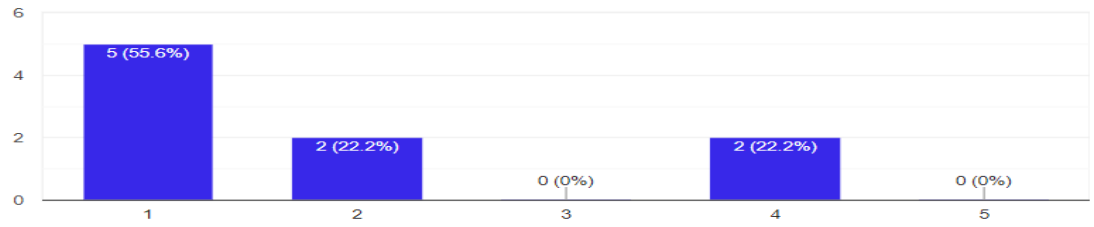
If you are the user of your phone on our website, what is your satisfaction with the design?

9 ردود



How easy to reach the books in System

9 ردود



What are the common mistakes in the system?

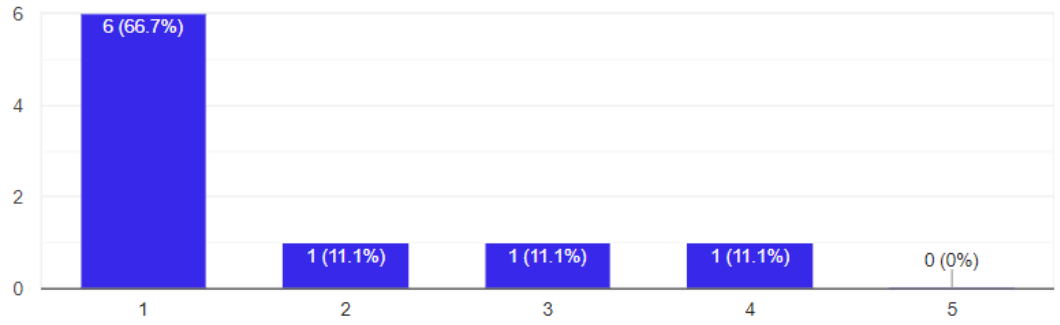
ردان (2)

it takes along time to download free books

Entering the code or the password wrong in the system

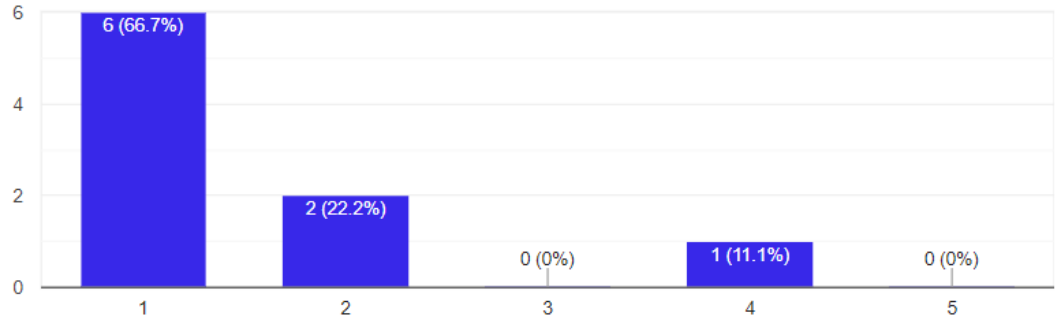
?" What is your satisfaction with the work of the website " system

9 ردود

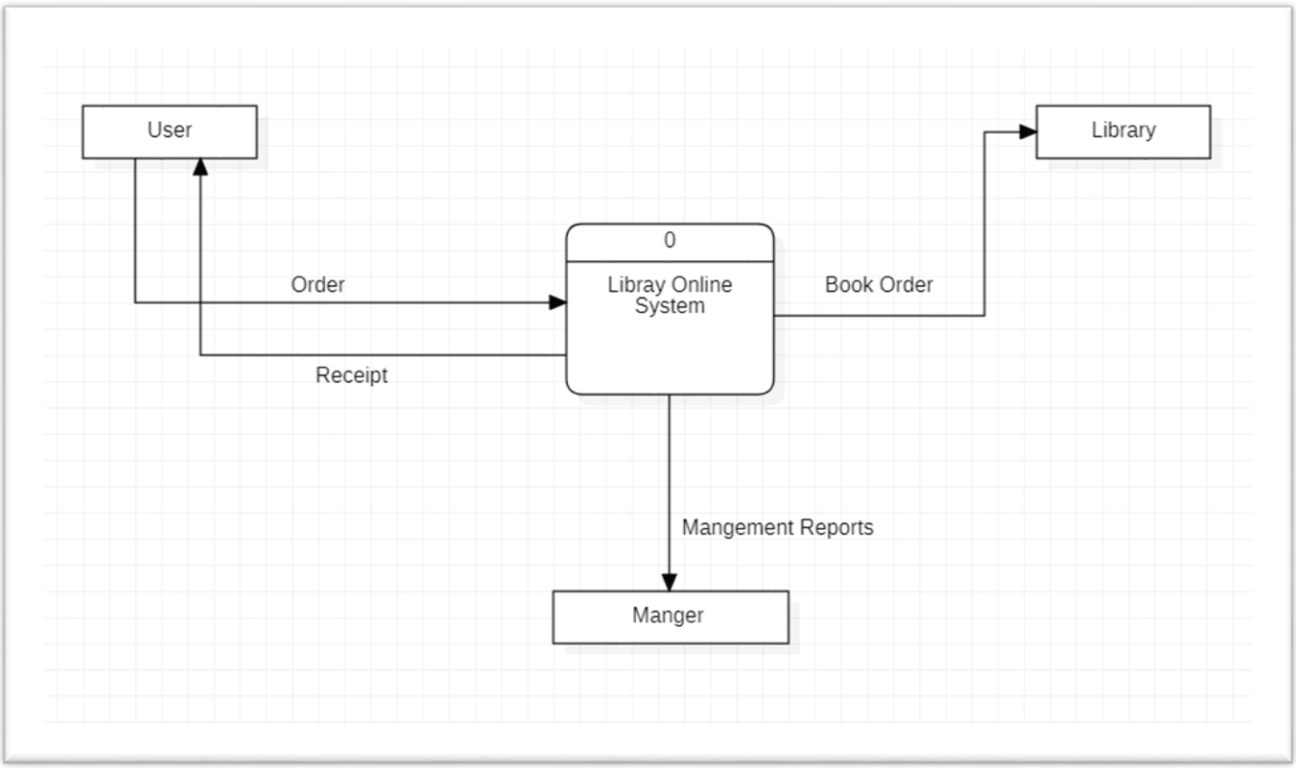


?Is the system easy to use

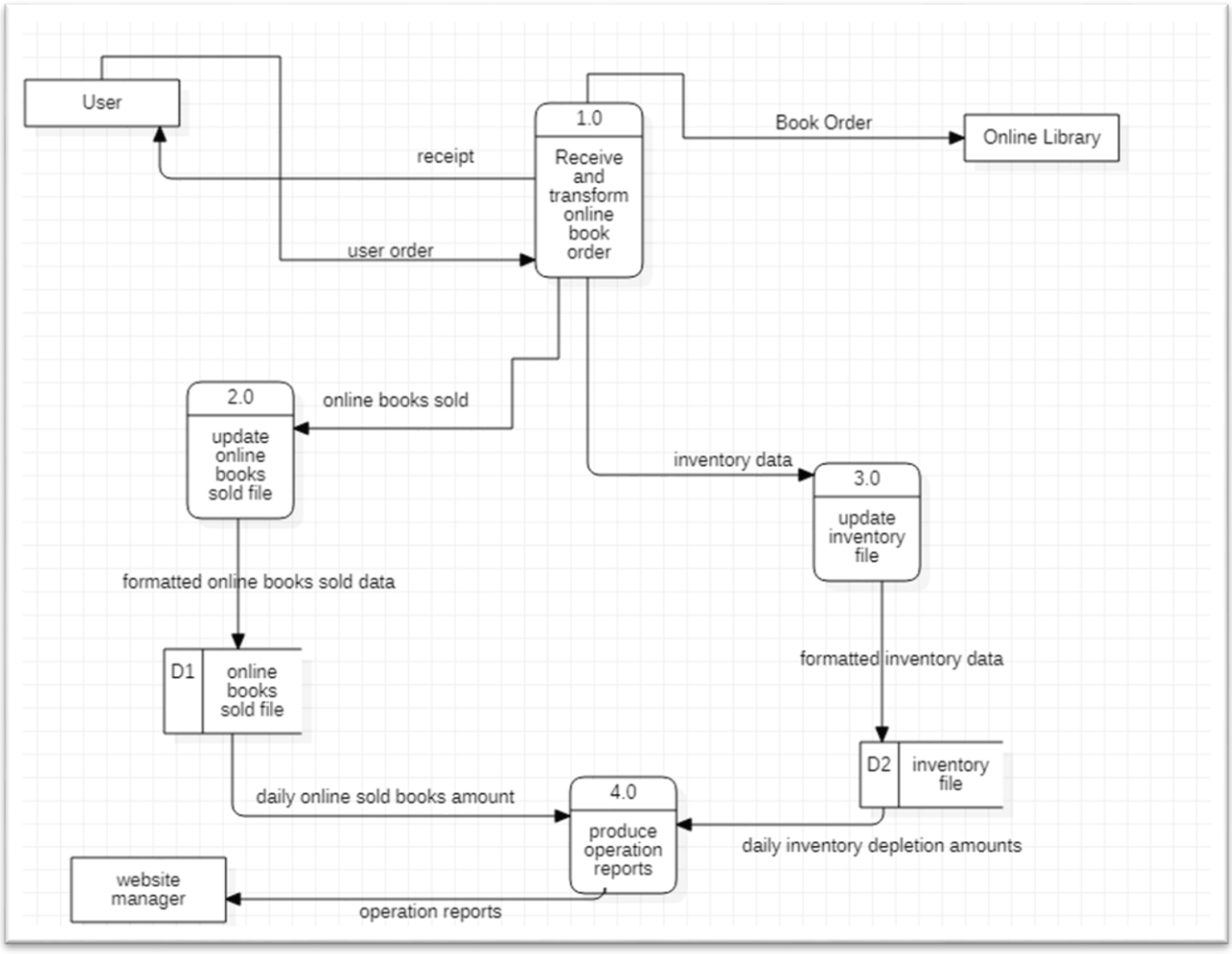
9 ردود



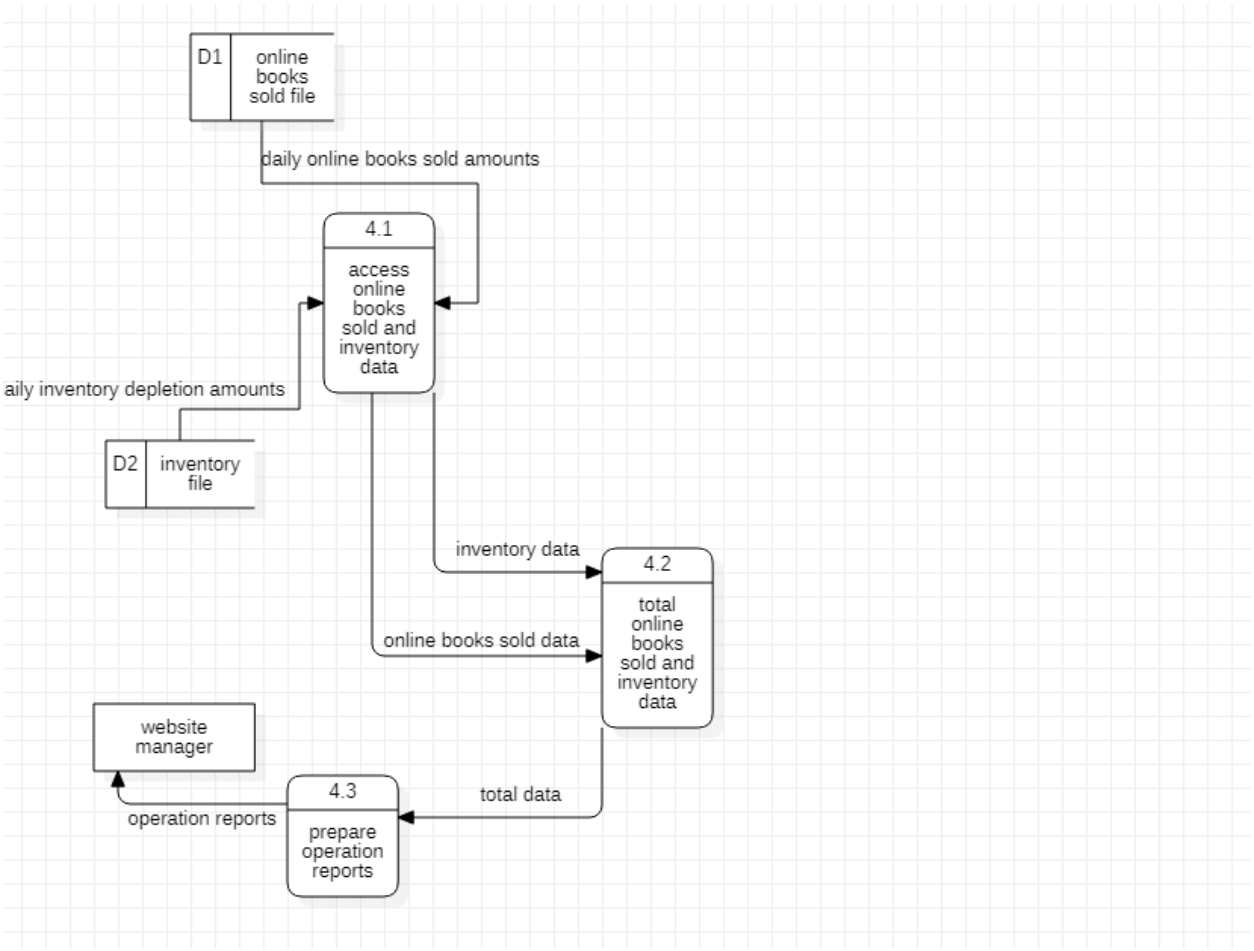
Data Flow Diagram - Context Diagram



Data Flow Diagram – Level/Diagram 0



Data Flow Diagram – Level 1



Data Dictionary

DFD: Data Flow

(1)Description form for Data Flow

Data Flow Description Form	
ID: 1 **Name (label): user order** **Description: contains the customer's order**	
Source **External entity - User**	**Destination** **Process 1.0 - receive and transform online book order**
Type of Data Flow **File** ☐ **Screen** ☐ **Report** ☐ **Form** ☐ **Internal** ☐	
Data structure **User data**	Volume/time **2/hour**
comments: **it contains the order of the customer about what the online book he need**	

Data Flow Description Form	
ID: 2 **Name (label): receipt** **Description: contains the receiving data**	
Source **Process 1.0 - receive and transform online book order**	**Destination** **External entity - User**
Type of Data Flow **File** ☐ **Screen** ☐ **Report** ☐ **Form** ☐ **Internal** ☐	
Data structure **Receipt data**	Volume/time **4/hour**
comments: **It contains the receiving data mean that the book orders which the customer request from the online library. It will appear a message which says if this book is existing or not.**	

Data Flow Description Form

ID: 3
Name (label): Book Order
Description: contains the transforming data

Source	**Destination**
Process 1.0 - receive and transform online book order	**External entity – online library**

Type of Data Flow

File ☐ **Screen** ☐ **Report** ☐ **Form** ☐ **Internal** ☑

Data structure	Volume/time
Transformation data	**3/hour**

comments:

it contains the transformation data about the book order after receiving it from the user

Data Flow Description Form

ID: 4
Name (label): online books sold
Description: contains the sold information data

Source	**Destination**
Process 1.0 - receive and transform online book order	**Process 2.0 – update online books sold file**

Type of Data Flow

File ☑ **Screen** ☐ **Report** ☐ **Form** ☐ **Internal** ☐

Data structure	Volume/time
Sold data	**8/hour**

comments:

it contains the information of the sold books like how many and what the amount, etc.

Data Flow Description Form

ID: 5
Name (label): inventory data
Description: contains information about the inventory

Source	**Destination**
Process 1.0 - receive and transform online book order	**Process 3.0 – update inventory file**

Type of Data Flow	
File ☑ Screen ☐ Report ☐ Form ☐ Internal ☐	
Data structure **Inventory data**	Volume/time **10/hour**
comments: **It contains the information about the inventory of the books because it needs to be updated every once the customer buy a book.**	

Data Flow Description Form

ID: 6
Name (label): formatted online books sold data
Description: contains the formatted sold information

Source	Destination
Process 2.0 – update online books sold file	**Data store – online books sold file**
Type of Data Flow **File** ☐ **Screen** ☐ **Report** ☐ **Form** ☐ **Internal** ☑	
Data structure **Formatted sold data**	Volume/time **9/hour**
comments: **it contains the information data about the sold online books after being formatted**	

Data Flow Description Form

ID: 7
Name (label): formatted inventory data
Description: contains the formatted inventory information

Source	Destination
Process 3.0 – update inventory file	**Data store – inventory file**
Type of Data Flow **File** ☐ **Screen** ☐ **Report** ☐ **Form** ☐ **Internal** ☑	
Data structure **Formatted inventory data**	Volume/time **9/hour**
comments:	

it contains the formatted information data about the inventory

Data Flow Description Form

ID: 8
Name (label): daily online sold books amount
Description: it contains the daily sold amount

Source	Destination
Data store – online books sold file	Process 4.0 – produce operation reports
Type of Data Flow File ☐ Screen ☐ Report ☐ Form ☐ Internal ☐	
Data structure Sold amount data	Volume/time 4/hour
comments: contains the online sold books amount report which been sold the whole day	

Data Flow Description Form

ID: 9
Name (label): daily inventory depletion amount
Description: contains the daily inventory amount

Source	Destination
Data store – inventory file	Process 4.0 – produce operation reports
Type of Data Flow File ☐ Screen ☐ Report ☐ Form ☐ Internal ☐	
Data structure Inventory amount data	Volume/time 4/hour
comments: it contains the information about the daily inventory amount data (the store)	

Data Flow Description Form

ID: 10
Name (label): operation reports
Description: contains the final report

Source	Destination
Process 4.0 – produce operation reports	External entity –website manager

Type of Data Flow File ☐ Screen ☐ Report ☐ Form ☐ Internal ☐	
Data structure **Report data**	Volume/time **2/hour**
comments: **It contains the final report data after the system of the online books library work the whole day. It will go to the website manager at the end**	

(2)Data Structure for Data Flow

User = first name + last name + password + phone number + user address + email + credit card number + security code

Receipt = first name + last name + phone number + user address + email + book name

Book order = book name + first name + last name + password + phone number + user address + email + credit card number + security code

Online books sold = book name + sold amount + [profits, losses] + first name + last name + book code + book ID

Inventory data = book name + inventory amount + book code + book ID

Formatted online books sold data = book name + book ID + book code + sold amount + [profits, losses] + first name + last name

Formatted inventory data = book name + inventory amount + book code + book ID

Daily online books sold amount = book name + book ID + book code + sold amount + [profits, losses] + first name + last name + user address + email + (date)

Daily inventory depletion amount = book name + inventory amount + book code + book ID + (date)

Operation reports = first name + last name + phone number + user address + email + book name + sold amount + [profits, losses] + book code + book ID + inventory amount + (date)

- **Data Structure for Data Flow: Structural Record**

User = first name + last name + password + phone number + user address + email + credit card number + security code

Receipt = first name + last name + phone number + user address + email + book name

Book order = book name + first name + last name + password + phone number + user address + email + credit card number + security code

Online books sold = book name + sold amount + [profits, losses] + first name + last name + book code + book ID

Inventory data = book name + inventory amount + book code + book ID

Formatted online books sold data = book name + book ID + book code + sold amount + [profits, losses] + first name + last name

Formatted inventory data = book name + inventory amount + book code + book ID

Daily online books sold amount = book name + book ID + book code + sold amount + [profits, losses] + first name + last name + user address + email + (date)

Daily inventory depletion amount = book name + inventory amount + book code + book ID + (date)

Operation reports = first name + last name + phone number + user address + email + book name + sold amount + [profits, losses] + book code + book ID + inventory amount + (date)

(2)Data Element for data flow

ID	
Name	**First name**
Alias	**Customer first name**
Description	**Contains the customer`s first name**
Length	**20**
Input Format	**X(20)**
Output Format	**X(20)**
Default Value	
Continuous / Discrete	**Discrete**
Type	**Alphabetic**
Base / Derived	**Base**
Upper Limit	
Lower Limit	
Discrete	

Comments	**Every customer has a first name**

ID	
Name	**Second name**
Alias	**Customer second name**
Description	**Contains the customer`s second name**
Length	**20**
Input Format	**X(20)**
Output Format	**X(20)**
Default Value	
Continuous / Discrete	**Discrete**
Type	**Alphabetic**
Base / Derived	**Base**
Upper Limit	
Lower Limit	
Discrete	
Comments	**Every customer has a second name**

ID	
Name	**Password**
Alias	**Customer password**
Description	**Contains the customer`s password**
Length	**10**
Input Format	**9(10)**
Output Format	**9(10)**
Default Value	
Continuous / Discrete	**Continuous**
Type	**Number**
Base / Derived	**Derived**
Upper Limit	
Lower Limit	
Discrete	
Comments	**Every customer should has a password**

ID	
Name	**Phone number**
Alias	**Customer phone number**
Description	**Contains the customer`s phone number**
Length	**15**
Input Format	**9(15)**
Output Format	**9(15)**
Default Value	

Continuous / Discrete	**Continuous**
Type	**Number**
Base / Derived	**Derived**
Upper Limit	
Lower Limit	
Discrete	
Comments	**Every customer should have a phone number**

ID	
Name	**User address**
Alias	**Customer address**
Description	**Contains the customer`s address**
Length	**30**
Input Format	**X(30)**
Output Format	**X(30)**
Default Value	
Continuous / Discrete	**Discrete**
Type	**Alphabet**
Base / Derived	**Base**
Upper Limit	
Lower Limit	
Discrete	
Comments	**Every customer should have an address**

ID	
Name	**Email**
Alias	**Customer email**
Description	**Contains the customer`s email**
Length	**40**
Input Format	**X(40)**
Output Format	**X(40)**
Default Value	
Continuous / Discrete	**Discrete**
Type	**Alphabet**
Base / Derived	**Base**
Upper Limit	
Lower Limit	
Discrete	
Comments	**Every customer should have address**

ID	
Name	**Credit card number**

Alias	**Customer credit card**
Description	**Contains the customer`s credit card number or code**
Length	**20**
Input Format	**9(20)**
Output Format	**9(20)**
Default Value	
Continuous / Discrete	**Continuous**
Type	**Number**
Base / Derived	**Derived**
Upper Limit	
Lower Limit	
Discrete	
Comments	**Every customer should have a credit card**

ID	
Name	**Security code**
Alias	**Customer security code**
Description	**Contains the customer`s security code**
Length	**30**
Input Format	**9(30)**
Output Format	**9(30)**
Default Value	
Continuous / Discrete	**Continuous**
Type	**Number**
Base / Derived	**Derived**
Upper Limit	
Lower Limit	
Discrete	
Comments	**Every customer should have a security code to keep his information safe**

ID	
Name	**Book name**
Alias	**Book`s name**
Description	**Contains the book`s name**
Length	**25**
Input Format	**X(25)**
Output Format	**X(25)**
Default Value	
Continuous / Discrete	**Discrete**
Type	**Alphabet**
Base / Derived	**Base**

Upper Limit	
Lower Limit	
Discrete	
Comments	**Every book should have a name**

ID	
Name	**Sold amount**
Alias	**Number of sold amount**
Description	**Contains the number of the sold amount**
Length	**10**
Input Format	**9(10)**
Output Format	**9(10)**
Default Value	
Continuous / Discrete	**Continuous**
Type	**Number**
Base / Derived	**Derived**
Upper Limit	
Lower Limit	
Discrete	
Comments	**The sold amount is important to know daily**

ID	
Name	**[profits, losses]**
Alias	**Profits or losses**
Description	**Contains the number of the profits or the losses**
Length	**10**
Input Format	**9(10)**
Output Format	**9(10)**
Default Value	
Continuous / Discrete	**Continuous**
Type	**Number**
Base / Derived	**Derived**
Upper Limit	
Lower Limit	
Discrete	
Comments	**We should always know the number of the profits or the losses**

ID	
Name	**Book code**
Alias	**Book code**
Description	**Contains the book`s code**

Length	**20**
Input Format	**9(20)**
Output Format	**9(20)**
Default Value	
Continuous / Discrete	**Continuous**
Type	**Number**
Base / Derived	**Derived**
Upper Limit	
Lower Limit	
Discrete	
Comments	**The book should have a code**

ID	
Name	**Book ID**
Alias	**Book ID**
Description	**Contains the book`s ID**
Length	**20**
Input Format	**9(20)**
Output Format	**9(20)**
Default Value	
Continuous / Discrete	**Continuous**
Type	**Number**
Base / Derived	**Derived**
Upper Limit	
Lower Limit	
Discrete	
Comments	**The book should has an ID**

ID	
Name	**Inventory amount**
Alias	**Inventory amount**
Description	**Contains the inventory`s amount**
Length	**15**
Input Format	**9(15)**
Output Format	**9(15)**
Default Value	
Continuous / Discrete	**Continuous**
Type	**Number**
Base / Derived	**Derived**
Upper Limit	
Lower Limit	
Discrete	
Comments	**Every day the inventory will be updated**

ID	
Name	**(date)**
Alias	**Date**
Description	**Contains the operation`s date**
Length	**20**
Input Format	**9(20)**
Output Format	**9(20)**
Default Value	
Continuous / Discrete	**Continuous**
Type	**Number**
Base / Derived	**Derived**
Upper Limit	
Lower Limit	
Discrete	
Comments	**The date of every operation is an optional element**

DFD: Data Store

ID	
Name	**Online books sold file**
Alias	**Sold file**
Description	**Contains the sold file**
File Type	**Database**
File Format	
Record Size	**20**
Maximum Records	**40**
Average Records	**30**
Growth per Year	**4/year**
Data Set	
Data Structure	**Store data**
Primary Key	
Secondary Key	
Comments	**The sold amount should be stored daily**

ID	
Name	**Inventory file**
Alias	**Inventory file**
Description	**Contains the inventory file**
File Type	**Database**
File Format	

Record Size	**20**
Maximum Records	**40**
Average Records	**30**
Growth per Year	**4/year**
Data Set	
Data Structure	**Store data**
Primary Key	
Secondary Key	
Comments	**The inventory amount are being updated daily so that it should be stored daily**

Process Specification

- ✓ Note: apply process specification for all project processes

Process Specification Form

Number: **1.o**

Name: **Receive and transform order**

Description: **This process use to receive and transform the user order**

Input Data Flow

The user order from External Entity

Output Data Flow

Send the order to the online library and receipt the order

Type of Process

Online **Batch** **Manual**

Process Logic:

IF customer is preferred customer then

If Customer order more than 1000$ then

Apply a 5% discount

If Customer use char card

Apply an additional 5% discount then

End if

Award 25$ bonus coupon

End if

Else Award 5$ bonus coupon

End if

Structured English **Decision Table** **Decision Tree**

Unresolved Issues:

Process Specification Form

Number: 2.o

Name: update online books sold file

Description: this process receive the update of online Sold Books and store the update in the online books Sold file

Input Data Flow :

the online sold books

Output Data Flow

Formatted online books sold file

Type of Process

Online **Batch** **Manual**

Process Logic:

		1	2	3	4	5	6	7	8
1	Lack of books	Y	Y	Y	Y	N	N	N	N
2	New book release	Y	Y	N	N	Y	Y	N	N
3	Laggy server	Y	N	Y	N	Y	N	Y	N

1	New update	X	X	X	X				
2	Add the new version	X	X			X	X		
3	New features	X		X		X		X	
4	Weekly update								X

Structured English Decision Table Decision Tree

Unresolved Issues:

Process Specification Form

Number: 3.o

Name: update inventory file

Description: this process receive the inventory data ,update it and formatted it into inventory file

Input Data Flow : inventory data

inventory data

Output Data Flow :

Formatted inventory data

Type of Process

Online Batch Manual

Process Logic:

		1	2	3	4	5	6	7	8
1	Lack of quantity	Y	Y	Y	Y	N	N	N	N
2	The order is greater than quantity	Y	Y	N	N	Y	Y	N	N
3	The quantity is not knowing	Y	N	Y	N	Y	N	Y	N

1	Update a new stock	X	X	X	X				
2	Move order to backordered record	X	X			X	X		
3	Do quantity inventory	X		X		X		X	
4	Weekly inventory								X

Structured English Decision Table Decision Tree

Unresolved Issues:

Process Specification Form

Number: 4.o

Name: produce operation reports

Description: this process receive the daily inventory depletion amounts and the daily online sold books amounts and do operation report and send it to the website manager

Input Data Flow :

daily online sold books amounts and inventory depletion amounts

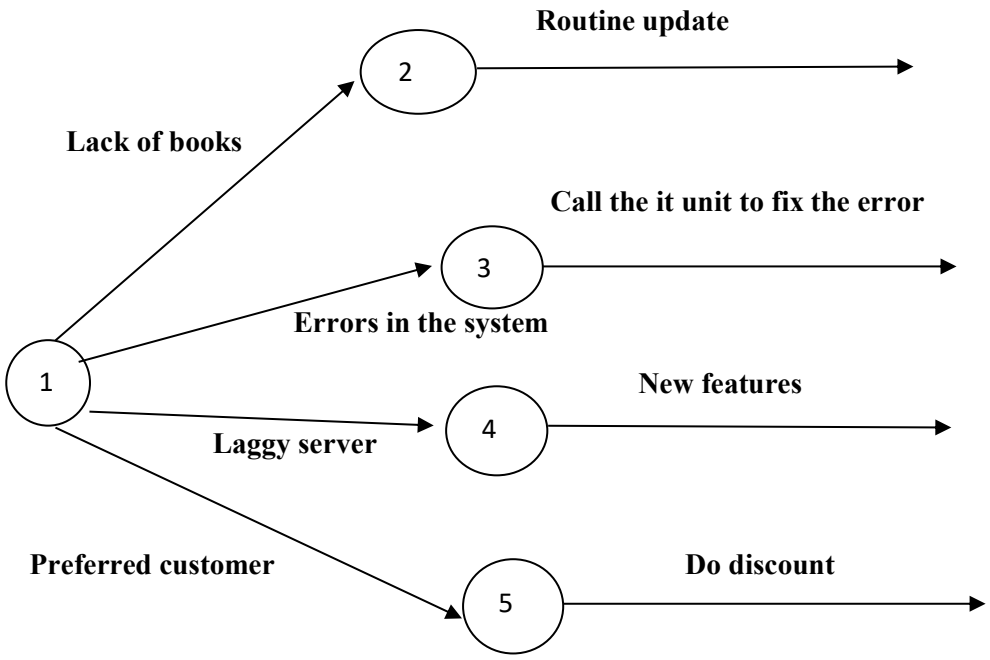
Output Data Flow

The operation report

Type of Process

Online **Batch** **Manual**

Process Logic:



Structured English **Decision Table** **Decision Tree**

Unresolved Issues:

Normalization

Normal Tables

id	Name	email	**Book id**	price	Payment type	**Publisher id**	Publisher name	address	
1	ahmed	Ah12@gmail.com	3	16$	visa	2	Hassan	Cairo	
2	ali	Al12@gmail.com	3	16$	visa	3	Osama	Giza	
3	rawan	ra12@gmail.com	4	20$	visa	1	Mohamed	Aswan	
1	ahmed	Ah12@gmail.com	4	10$	cash	3	Osama	Giza	
3	rawan	ra12@gmail.com	2	10$	cash	2	Hassan	Cairo	
4	joulnar	jo18@gmail.com	4	20$	visa	4	Jone	USA	
2	ali	Al12@gmail.com	3	16$	visa	1	Mohamed	Aswan	

Tables after 1NF

id	Name	email
1	ahmed	Ah12@gmail.com
2	Ali	Al12@gmail.com
3	Rawan	ra12@gmail.com
4	joulnar	jo18@gmail.com

id	**Book id**	price	Payment type	**Publisher id**	Publisher name	address
1	3	16$	visa	2	Hassan	Cairo
2	3	16$	visa	3	Osama	Giza
3	4	20$	visa	1	Mohamed	Aswan
1	4	10$	cash	3	Osama	Giza
3	2	10$	cash	2	Hassan	Cairo
4	4	20$	visa	4	Jone	USA
2	3	16$	visa	1	Mohamed	Aswan

Tables after 2NF

id	Name	email
1	ahmed	Ah12@gmail.com
2	Ali	Al12@gmail.com
3	Rawan	ra12@gmail.com
4	joulnar	jo18@gmail.com

Publisher id	Publisher name	address
2	Hassan	Cairo
3	Osama	Giza
1	Mohamed	Aswan
4	Jone	USA

id	**Publisher id**	**Book id**	price	Payment type
1	2	3	16$	visa
2	3	3	16$	visa
3	1	4	20$	visa
1	3	4	10$	cash
3	2	2	10$	cash
4	4	4	20$	visa
2	1	3	16$	visa

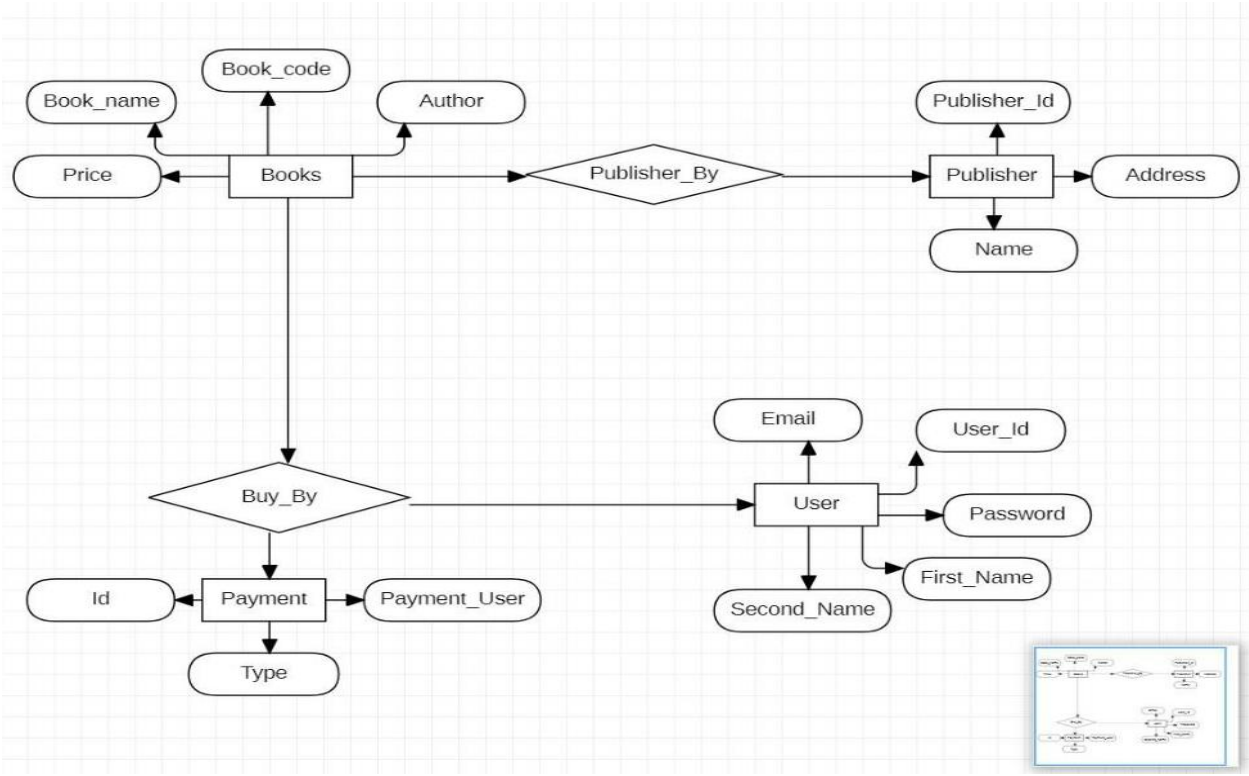
Tables after 3NF

id	Name	email
1	ahmed	Ah12@gmail.com
2	Ali	Al12@gmail.com
3	Rawan	ra12@gmail.com
4	joulnar	jo18@gmail.com

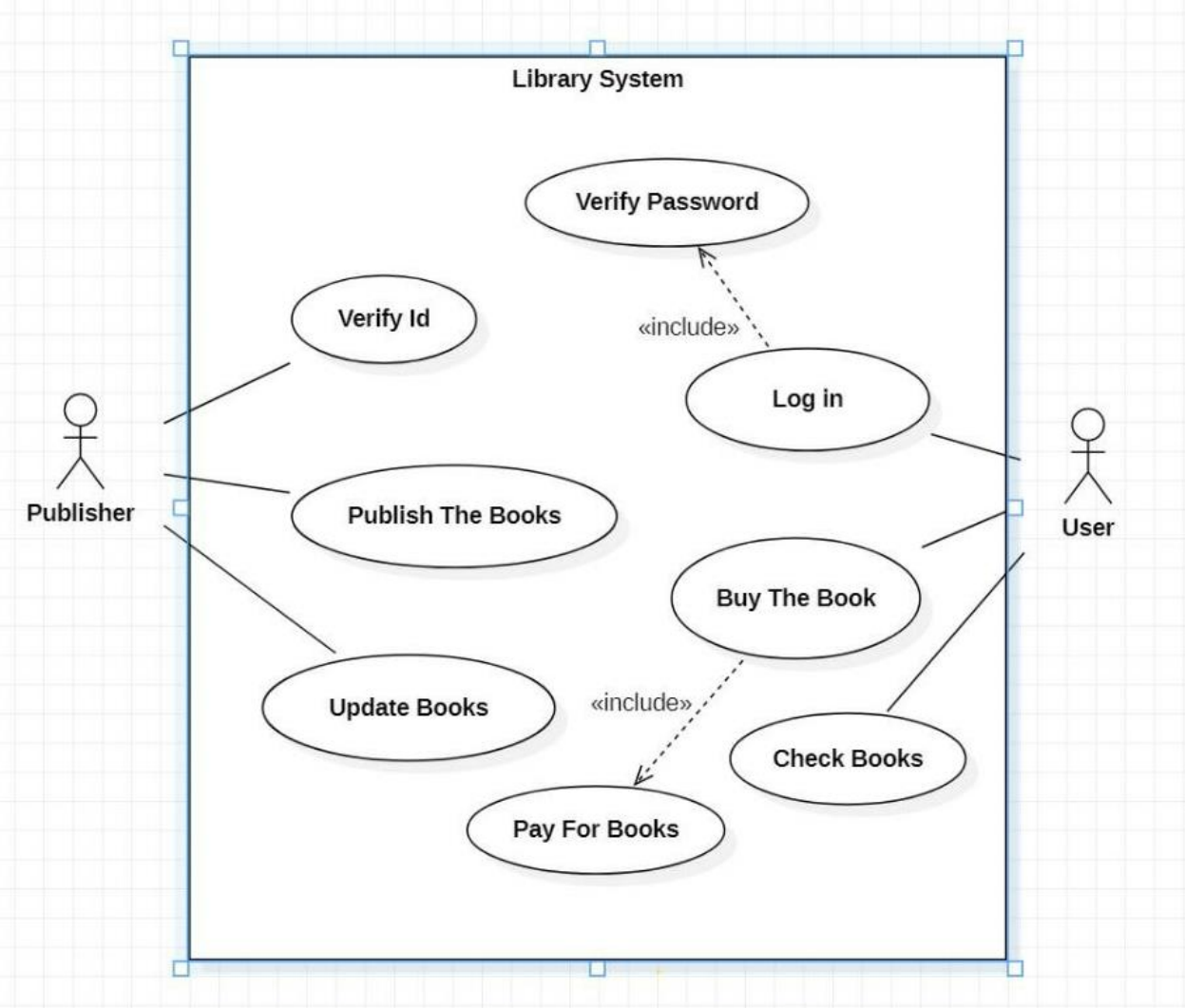
Publisher id	Publisher name	address
2	Hassan	Cairo
3	Osama	Giza
1	Mohamed	Aswan
4	Jone	USA

id	Publisher id	Book id	price	Payment type
1	2	3	16$	visa
2	3	3	16$	visa
3	1	4	20$	visa
1	3	4	10$	cash
3	2	2	10$	cash
4	4	4	20$	visa
2	1	3	16$	visa

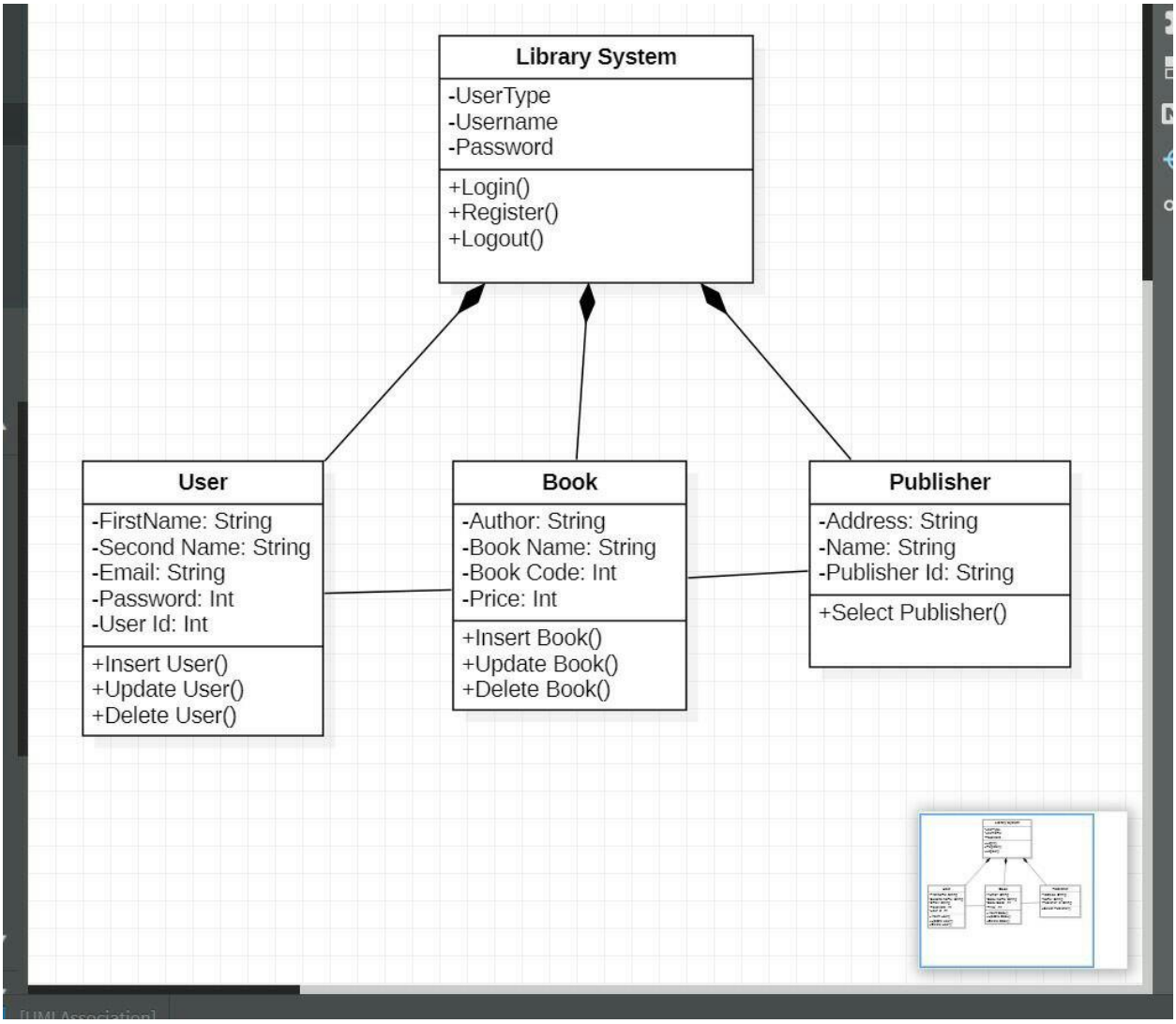
Entity Relationship Diagram (ERD)



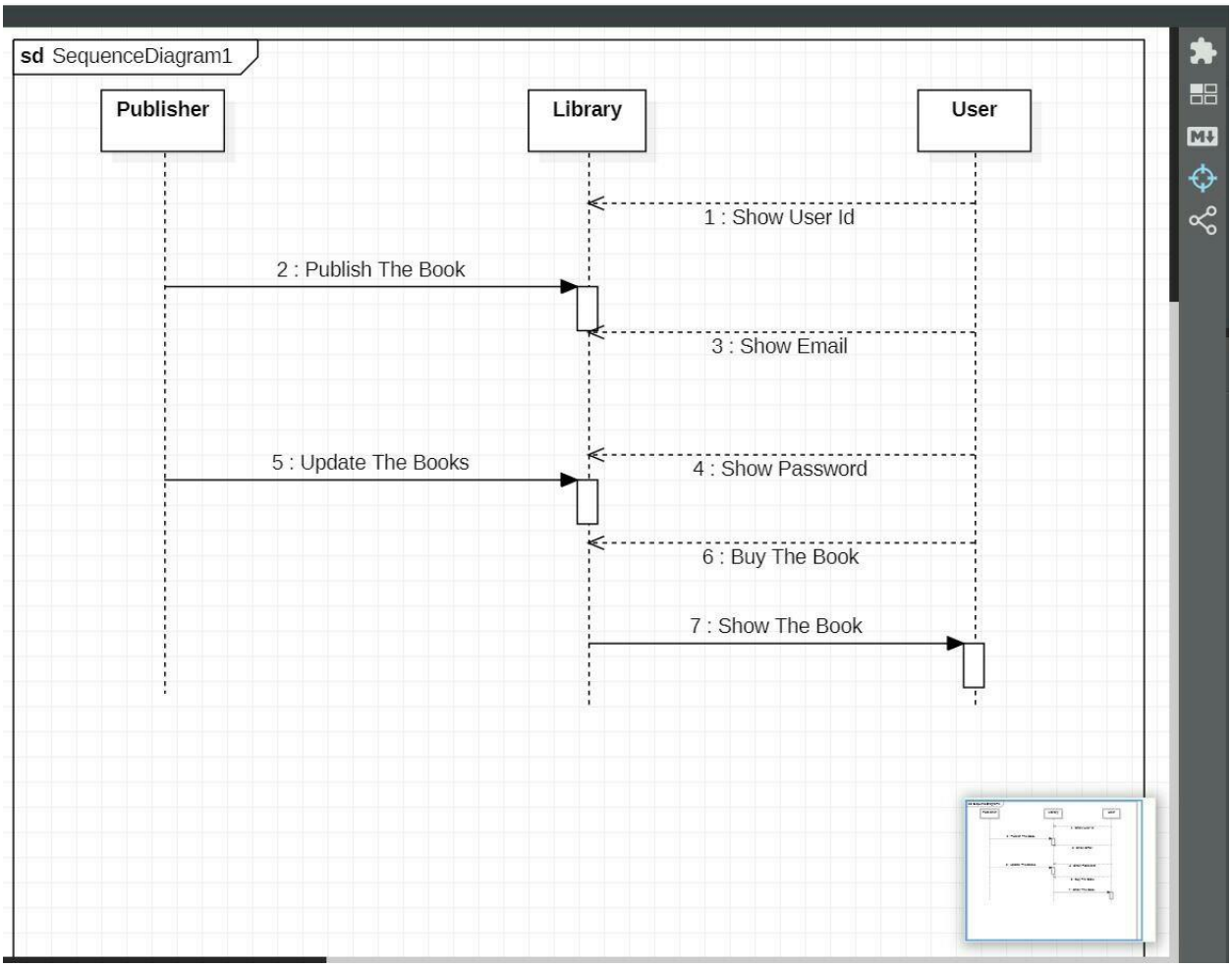
UML – Use Case


Library System
Verify Password
Verify Id
«include»
Log in
Publisher
Publish The Books
User
Buy The Book
Update Books
«include»
Check Books
Pay For Books

UML – Class Diagram



UML – Sequence Diagram



Design (Project Specification)

[A] Architecture Design

1. Operational Requirements

Requirement	
Technical Environment	
System Integration	
Portability	
Maintainability	

2. Performance Requirements

Requirement	
Speed	
Capacity	
Availability and reliability	

3. Security Requirements

Requirement	
System value estimates	
Access control	
Encryption and authentication	
Virus control	

[2] Data storage Design

1. Data storage format:

2. Convert logical ERD to physical ERD:

[3] Program design

- **Convert logical DFD to physical DFD:**

[4] Interface design

1. Use Scenario Development

2. Interface Structure Design

3. Interface Standards Design

(1) Interface Metaphor

(2) Interface Template

4. Interface Design Prototyping

Implementation (New System)